Hilbert's 17th Problem and the Pandrosion Sum of Squares

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Abstract

Hilbert's 17th problem asks whether every non-negative polynomial is a sum of squares of rational functions. Artin answered affirmatively in 1927. In the univariate case, the Pandrosion Turán identity $P'^2 - PP'' = \sum Q(\alpha_j, x)^2$ provides a *constructive polynomial* SOS decomposition with exactly n summands—one per root—that is both existence-free and optimal. We extend this to separable multivariate polynomials via the partial Turán, verify the Motzkin polynomial as a negative control (non-negative but not polynomial SOS), and compare the Pandrosion decomposition to generic SOS bounds.

1 Hilbert's 17th problem

Theorem 1.1 (Artin, 1927)**.** *Every polynomial $f \in \mathbb{R}[x_1, \dots, x_m]$ satisfying $f \geq 0$ on $\mathbb{R}^m$ is a sum of squares of* rational functions*: $f = \sum (g_i/h_i)^2$ for $g_i, h_i \in \mathbb{R}[x_1, \dots, x_m]$.*

Theorem 1.2 (Hilbert, 1888)**.** *A non-negative polynomial is a sum of squares of* polynomials *if and only if $(m, 2d)$ falls in one of three cases: (a) any m, degree 2; (b) $m = 1$, any degree; (c) $m = 2$, degree 4.*

2 The Pandrosion answer: Turán SOS

Theorem 2.1 (Pandrosion SOS, Paper 56)**.** *For $P(x) = \prod_{j=1}^{n}(x - \alpha_j)$ with all $\alpha_j \in \mathbb{R}$:*

$$\boxed{P'(x)^2 - P(x)\,P''(x) = \sum_{j=1}^{n} Q(\alpha_j, x)^2,} \tag{1}$$

where $Q(\alpha_j, x) = \prod_{k \neq j}(x - \alpha_k)$. This is a sum of ***exactly*** *n* ***polynomial squares****, each of degree $n - 1$.*

This is a constructive, existence-free instance of case (b) in Theorem 1.2: the Turán form of a real-rooted polynomial of degree $2(n-1)$ is decomposed into n squares without any optimization or search.

n	$\deg(T)$	**Pandrosion: n squares**	**Generic bound**
3	4	3	9
5	8	5	25
10	18	10	100
20	38	20	400
50	98	50	2500

The Pandrosion decomposition requires exactly n squares versus $O(n^2)$ for generic SOS methods.

3 The Motzkin counterexample

Proposition 3.1. *The Motzkin polynomial $M(x,y) = x^4y^2 + x^2y^4 - 3x^2y^2 + 1$ satisfies $M \geq 0$ on $\mathbb{R}^2$ (with $M(\pm 1, \pm 1) = 0$), but M is **not** a sum of squares of polynomials.*

Verified numerically: $\min_{[-2,2]^2} M = 0.000$ (at $(\pm 1, \pm 1)$) with $M > 0$ elsewhere. This falls outside Hilbert's classification as $(m, 2d) = (2, 6)$.

4 Multivariate partial Turán

Proposition 4.1 (Partial Pandrosion SOS)**.** *For a separable polynomial $P(x,y) = P_x(x) \cdot P_y(y)$:*

$$\left(\frac{\partial P}{\partial x}\right)^2 - P \cdot \frac{\partial^2 P}{\partial x^2} = P_y(y)^2 \cdot \sum_j Q_x(\alpha_j, x)^2. \tag{2}$$

This is a polynomial SOS in both *x and y simultaneously.*

Verified: $P_x = (x+1)(x-1)(x-2)$, $P_y = y(y-3)$, 2500 grid points, 0 violations.

5 Pandrosion and Hilbert's programme

Hilbert 17 aspect	**Pandrosion tool**
Existence (Artin)	Constructive: $T = \sum Q_j^2$
Polynomial SOS (case b)	Turán form is always poly SOS
Number of squares	Exactly n (one per root)
Multivariate extension	Partial Turán: $P_y^2 \cdot \sum Q_x^2$
Non-negativity certificate	$\sum Q_j^2 \geq 0$ automatic
Motzkin obstruction	Outside Pandrosion class (no real roots)

References

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